

Johnny Tian-Zheng Wei

200 S El Molino St
Los Angeles, CA 91801

(323) 459-9836
jtwei@usc.edu
johntzwei.github.io

Education

Ph.D. Candidate in CS, *University of Southern California* Expected graduation: May 2026
Advisor: Robin Jia. Thesis: Statistically principled measurement of LLMs by spiking the training data.
Research interests: AI measurement, spiking, accountability, and safety.

B.S. in Mathematics, *University of Massachusetts Amherst*
GPA: 4.0 - Advisor: Brendan O'Connor. Thesis: Grammaticality in neural natural language generation.
Mathematics major with concentration in Statistics. Minors in Linguistics and Computer Science.

Research Experience

Allegro Lab, *University of Southern California* Aug. 2019 - Present

- *ICLR 2026*. Awarded a compute grant for 200,000 A100 hours ($\approx$ \$1M) to train the HUBBLE models for the study of LLM memorization in the context of copyright, privacy, and test set contamination.
- *FAccT 2025*. We discuss how fair use can evolve for LLMs and operationalize a fair learning doctrine based on model memorization. We further measure LLM memorization w.r.t. their design decisions.
- *ACL 2024 Findings*. To support a right to opt-out of LLM training, we propose the use of data watermarks which can detect whether your data was used for pretraining an LLM with statistical guarantees.
- Managed the USC NLP cluster of 80 A6000s. The cluster spans 5 research groups with 40+ active users. Installed a provisioning system, diagnosed hardware bottlenecks, and strengthened cluster governance.

Microsoft Research, *Remote work* May 2022 - Aug 2022

- Manager: Tom Kocmi. Proposed interim testing for evaluation. Interim testing allows for evaluations to stop early if a result can be concluded. By simulating on historical data, we could determine that 20% of the budget could be saved, while keeping the same sensitivity (on an annual \$2.5M+ budget).

Selected Publications

A full list of my publications can be found on Google Scholar.

Johnny Tian-Zheng Wei*, Jerry Li*, Ameya Godbole, Robin Jia. Spiking the training data to adjust for test set contamination. In submission to *COLM*.

Johnny Tian-Zheng Wei*, Ameya Godbole*, Mohammad Aflah Khan*, Ryan Wang, Xiaoyuan Zhu, James Flemings, Nitya Kashyap, Krishna P. Gummadi, Willie Neiswanger, Robin Jia. Hubble: a model suite to advance the study of LLM memorization. In the *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026.

Johnny Tian-Zheng Wei*, Maggie Wang*, Ameya Godbole, Jonathan H. Choi, Robin Jia. Interrogating LLM design under a fair learning doctrine. In the *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2025.

Johnny Tian-Zheng Wei*, Ryan Wang*, Robin Jia. Proving membership in LLM pretraining data via data watermarks. In the *Findings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

Johnny Tian-Zheng Wei, Frederike Zufall, Robin Jia. Operationalizing content moderation “accuracy” in the Digital Services Act. In the *AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2024.

Teaching Experience

Teaching Assistant, *University of Southern California* Sept. 2020 - Present

- CSCI 270 Introduction to Algorithms and Theory of Computing, Spring 2025.
- CSCI 699 The Science of Large Language Models, Fall 2024.
- CSCI 567 Machine Learning, Summer 2021.
- CSCI 103 Introduction to Programming, Fall 2020, Spring 2021.

Service

- Organizer for the First Workshop on Large Language Model Memorization (L2M2) at ACL 2025.
- Reviewer for ICLR 2026, FAccT 2026, 2025, 2024, ACL 2025, 2024, 2021.